Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/15/2014
PI Name:	Rubin, David M.S.		
Project Title:	Middleware for Exploration Medical System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	david.a.rubin@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281.461.2728
Organization Name:	Wyle Laboratories, Inc./NASA Johnson Space Center		
PI Address 1:	Project Engineer/ISS Biomedical Flight Controller		
PI Address 2:	1290 Hercules Avenue		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2011	**End Date:**	03/27/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Watkins, Sharmi1a	**Contact Phone:**	281.483.0395
Contact Email:	sharmila.watkins@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/27/2015 (from 9/30/2018) per PI (Ed., 6/30/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

The middleware will be intelligent data management and communications software that facilitates the acquisition, integration, management and distribution of data from on-orbit vehicle medical and environmental devices, sensors, and systems. The software will reduce crew time and ensure the accuracy of data collection and timeliness of data delivery. Development of the middleware addresses the following risk as stated within the Critical Path Roadmap (CPR; also known as the NASA Bioastronautics Roadmap (NASA/SP-2004-6113)):

Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities: Given that medical capabilities will be limited during human exploration missions, there is a possibility that in-flight medical events will lead to unacceptable health and mission outcomes.

The objectives of the middleware are to:

Task Description:	a. Reduce or eliminate the time required of an on-orbit crew and ground personnel [which include Surgeon, Biomedical Engineer (BME), and Medical Operations Data Specialist] to access, transfer, and manipulate medical data. b. Support the integrated demonstration that the on-orbit crew has the ability to access medical data/information via an intuitive and crew-friendly solution to assist/aid in the treatment of a medical condition. c. Develop a common data management framework and architecture that can be ubiquitously used to automate repetitive data collection, management, and communications tasks for all crew health and life sciences activities. d. Ensure crew access to medical data during periods of reduced or unavailable ground communication. e. Develop a common data management framework and architecture that allows for scalability, extensibility, and interoperability of data sources and data users. f. Lower total cost of ownership for development and sustainment of peripheral hardware and software that utilize the Middleware for data management g. Provide better crew health via reduction in crew errors, crew time, and ground time. The ground demonstration of the middleware, as part of the Exploration Medical System Demonstration (EMSD), will occur in 2015 and the in-flight phase will take place onboard the ISS in 2018.
Rationale for HRP Directed Research:	This study is a combination of a solicited task (RFP released under Wyle for procurement of COTS middleware system) and a directed task involving customization of the software for the specific needs of ExMC. The study team is uniquely positioned to perform this function and to ensure that the middleware component of the overall Exploration Medical System Demonstration (EMSD) is integrated with the other components of the system.
Research Impact/Earth Benefits:	The Exploration Medical System is being designed specifically for a Near Earth Asteroid mission and, as such, has no direct impact or benefits to life on Earth. However, the middleware system, as part of the larger Exploration Medical System, has potential training and simulation opportunities for training of future astronauts and ground personnel. In addition, the Exploration Medical System can serve as software infrastructure for future development and integration of medical devices and software.
Task Progress:	The following accomplishments have been made during the current reporting period of 6/30/13-7/1/14: -The EMSD team participated in the ExMC System Requirements Review (Phase 3) by submitted updates to the ExMC Operational Concepts (OpsCon) document and Functional Requirements Document (FRD), including a new section for the image and video capture system of EMSD. The EMSD team also participated in the RID review and disposition process for these documents. -The EMSD team procured and installed the middleware software on the test computing platform. -The EMSD team conducted the Detailed Design Phase of the project. As part of this phase, the EMSD team conducted the Critical Design Review Kick Off and RID Board meetings. All RIDs were dispositioned and implemented and all project documents were updated accordingly. There were no impacts to the middleware or integration of the middleware into the EMSD system. -The EMSD team developed a Validation and Verification Document which will be used, in part, to validate and verify middleware functionality within the EMSD system. -The EMSD team began the Production, Test, and Certification Phase of the EMSD project. This includes the development of additional middleware functionality, testing of that functionality, and integration of the middleware with the other EMSD components. -The EMSD team integrated the NSBRI SpaceMED system into the EMSD system. This may include the transmittal of data, produced by the SpaceMED system, with the assistance of the EMSD middleware component.
Bibliography Type:	Description: (Last Updated: 04/16/2014)
Abstracts for Journals and Proceedings	Rubin DA, Watkins SD. "Exploration Medical System Demonstration Project." Presented at the 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3047.pdf , Feb-2014
Abstracts for Journals and Proceedings	Shah RV, Rubin DA, Kerstman EL. "NASA Exploration Medical Capability Exploration Medical System Demonstration Project." Presented at the 85th Annual Scientific Meeting, Aerospace Medical Association, San Diego, CA, May 10-15, 2014. Aviation, Space, and Environmental Medicine. 2014 Mar;85(3):327. http://www.ingentaconnect.com/content/asma/asem/2014/00000085/00000003 , Mar-2014